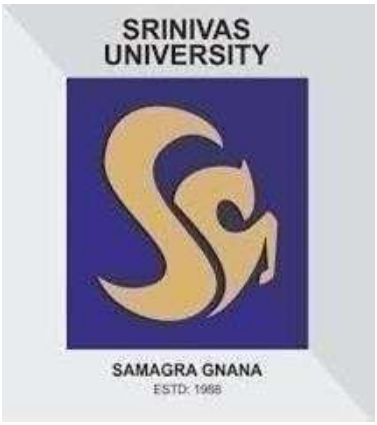
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Question Bank Answers with Bloom's Model

CLOUD SECURITY IN MICROSOFT AZURE ENVIRONMENT



MODULE 1

INTRODUCTION TO CLOUD SECURITY

Objective Type Questions (1 marks)

1. What is Identity and Access Management (IAM)?

A) Data Backup

B) Identity Verification and Access Control

C) Data Compression

D) Network Expansion

Answer: B) Identity Verification and Access Control

2. Which Azure service is used to manage user identities and authentication?

A) Azure Backup

B) Azure Monitor

C) Microsoft Entra ID

D) Azure Site Recovery

Answer: C) Microsoft Entra ID

3. What is the security technique that converts readable data into an unreadable format?

A) Authentication

B) Encryption

C) Monitoring

D) Auditing

Answer: B) Encryption

4. Which entity is responsible for managing physical servers in Azure under the Shared Responsibility Model?

A) Customer

B) End User

C) Microsoft Azure

D) Application Developer

Answer: C) Microsoft Azure

5. Which data protection regulation is designed to safeguard the personal data and privacy of individuals in the European Union (EU)?

A) HIPAA

B) PCI-DSS

C) GDPR

D) ISO 9001

Answer: C) GDPR

6. Which cloud service model is fully managed by the cloud provider?

A) IaaS

B) PaaS

C) SaaS

D) Hybrid Cloud

Answer: C) SaaS

7. Which entity is responsible for managing user accounts and passwords in Azure?

A) Microsoft Azure

B) Hardware Vendor

C) Customer Organization

D) Internet Service Provider

Answer: C) Customer Organization

8. Which compliance standard is related to Information Security Management Systems (ISMS)?

A) ISO 27001

B) ISO 9001

C) GDPR

D) HIPAA

Answer: A) ISO 27001

9. What is cloud security?

A) Managing cloud service costs

B) Protecting cloud data and resources from threats

C) Developing software applications

D) Increasing internet speed

Answer: B) Protecting cloud data and resources from threats

10. Which Azure service provides protection against DDoS attacks?

A) Azure Backup

B) Azure DDoS Protection

C) Azure SQL Database

D) Azure DevOps

Answer: B) Azure DDoS Protection

11. What approach would you use to protect cloud resources from unauthorized access?

A) Data compression

B) Access control

C) File sharing

D) Data duplication

Answer: B) Access control

12. How would you protect a web application from a large number of malicious requests?

A) Enable Azure DDoS Protection

B) Delete the application

C) Disable network security

D) Share the admin password

Answer: A) Enable Azure DDoS Protection

13. What action would you take if a storage account is accidentally made public?

A) Increase storage capacity

B) Disable public access

C) Create another subscription

D) Ignore the issue

Answer: B) Disable public access

14. What approach would you use to protect sensitive customer data from unauthorized access when it is stored in the cloud?

A) Share it with all users

B) Remove access controls

C) Encrypt the data

D) Make the data public

Answer: C) Encrypt the data

15. What would you do after detecting unusual login activity?

A) Give the user more access

B) Ignore the activity

C) Investigate and block suspicious access

D) Disable all security logs

Answer: C) Investigate and block suspicious access

16. How would you secure an Azure virtual machine from unauthorized network traffic?

A) By storing passwords in Azure Key Vault

B) By enabling Multi-Factor Authentication

C) By configuring a Network Security Group (NSG)

D) By monitoring application performance

Answer: C) By configuring a Network Security Group (NSG)

17. What approach would you use to recover data after a cloud security incident?

A) Delete all backups

B) Use a backup copy

C) Disable recovery services

D) Remove the affected data

Answer: B) Use a backup copy

18. What approach would you use to give a student access only to a specific cloud application?

A) Full administrator access

B) Access to all cloud resources

C) Only the required application access

D) Access to all user accounts

Answer: C) Only the required application access

19. What approach would you use to support GDPR requirements when handling customer data?

A) Collect unnecessary data

B) Share data publicly

C) Protect personal data and collect only what is needed

D) Keep all data without restrictions

Answer: C) Protect personal data and collect only what is needed

20. What would you do when an employee leaves the organization to prevent further access?

A) Keep the account active

B) Disable the user's access

C) Share the account with another employee

D) Give the account administrator rights

Answer: B) Disable the user's access

Short Answer Questions (3 Marks)

21. What is General Data Protection Regulation (GDPR)? Describe with an example.

Answer:

The General Data Protection Regulation (GDPR) is a law introduced by the European Union (EU) to protect people's personal data and privacy. It ensures that organizations collect, use, and store personal information safely and securely.

Example:

An Indian shopping website sells products to customers in Germany and stores their data in Microsoft Azure. Since it collects the personal data of EU customers, the company must follow GDPR rules.

22. Define Multi-Factor Authentication (MFA) with an example.

Answer:

Multi-Factor Authentication (MFA) is a security method that requires users to verify their identity using two or more authentication factors before accessing a system or application. It provides an extra layer of protection because a stolen or guessed password alone isn't enough to get in the attacker would also need the second factor, like your phone or fingerprint.

Example: An employee logs into Azure using a password and then confirms their identity through an OTP sent to their mobile phone.

23. What is a DDoS attack? Describe with an example.

Answer:

A Distributed Denial-of-Service (DDoS) attack is a cyberattack in which multiple compromised devices send a large amount of traffic to a target server, website, or network. This overloads the system, making it slow or unavailable to legitimate users.

Example:

A hacker sends a huge number of fake requests from many computers to an online shopping website. This makes the website slow or crash, so real customers cannot access it.

24. What is Role-Based Access Control (RBAC)? State its purpose.

Answer:

RBAC allows organizations to assign permissions based on user roles. Users receive only the access required to perform their job responsibilities.

Purpose of RBAC:

- Ensures users receive only the permissions required for their job (Principle of Least Privilege).
- Prevents unauthorized access to resources.
- Improves security and supports compliance.

25. **List the responsibilities of Microsoft Azure under the Shared Responsibility Model.**

Answer:

Microsoft Azure Responsibilities:

- Secures the physical data centers, servers, storage, and networking.

- Maintains the Azure infrastructure, hardware, and virtualization layer.
- Ensures the availability and security of cloud services.

26. Define ISO 27001 with an example.

Answer:

ISO/IEC 27001 is an international standard for establishing, implementing, and managing an Information Security Management System (ISMS) to protect sensitive information. It helps organizations identify security risks and ensure the confidentiality, integrity, and availability of information.

Example : A hospital follows ISO 27001 by using strong passwords, encryption, multi-factor authentication, regular backups, and access control to protect patient records.

27. List the responsibilities of customers under the Shared Responsibility Model.

Answer:

Customer Responsibilities:

- Manages user identities, passwords, and access permissions.
- Protects applications, operating systems (in IaaS), and customer data.
- Configures security settings, updates software, and follows security best practices.

28. Describe the Azure Shared Responsibility Model.

Answer:

Security responsibilities are shared between Microsoft and the customer.

- Microsoft secures the cloud infrastructure, hardware, networking, and data centers.
- Customers secure their data, applications, user accounts, and access controls.
- Azure provides security tools such as Microsoft Entra ID, RBAC, Azure Key Vault, and Defender for Cloud.
- Responsibility varies based on IaaS, PaaS, and SaaS service models.

29. List the components of the Azure Cloud Security Framework.

Answer:

1. **Identity and Access Management (IAM)**
2. **Data Encryption**
3. **Monitoring and Threat Detection**

4. **Network Security and Firewalls**
5. **Backup and Disaster Recovery**

30. Define Data Encryption. Describe its importance with an example.

Answer:

Data Encryption converts readable data into an unreadable format to protect it from unauthorized access.

Example: A bank stores customer account details in Azure Storage. The data is encrypted so that even if an attacker gains access to the storage, the information cannot be read without the proper decryption key.

Long Answer Questions (5 Marks)

31. List the three cloud service models and explain each with a suitable example.

Answer:

1. IaaS (Infrastructure as a Service)

In IaaS**,** Azure provides virtual machines, storage, and networking. The customer manages the operating system, applications, and security.

Example: A bank uses an Azure Virtual Machine to run its banking application.

2. PaaS (Platform as a Service)

In PaaS**,** Azure manages the infrastructure, operating system, and platform. The customer develops and manages the application and its data.

Example: College develops an online attendance and examination management system.

3. SaaS (Software as a Service)

In SaaS**,** Azure manages the application, infrastructure, updates, and maintenance. The customer only uses the software and manages user accounts and access.

Example: An organization uses Microsoft 365 for email, Word, Excel, and other office applications.

32. What is a Data Breach? Describe its impacts with an example.

Answer:

A data breach in cloud platforms like Microsoft Azure happens when an unauthorized person gains access to sensitive or confidential information stored in the cloud.

Impact of Data Breaches

1. Sensitive customer and business information.

2. Leading to financial losses, legal penalties.
3. Identity theft, Business disruption, and reputational damage.
4. Customers may lose trust in the organization, which can affect long-term business growth & operations.

33. Describe the Azure Security Framework Components.

Answer:

1. Identity and Access Management (IAM)

IAM controls who can access Azure resources. Azure uses Microsoft Entra ID with MFA, RBAC, SSO, and Conditional Access to provide secure access.

2. Data Encryption

Data Encryption converts readable data into an unreadable format to keep it safe.

3. Monitoring and Threat Detection

Monitoring and Threat Detection continuously checks Azure resources for security threats. It detects suspicious activities and sends alerts to administrators.

4. Network Security and Firewalls

Network Security and Firewalls protect Azure resources from hackers and unwanted network traffic

5. Backup and Disaster Recovery

It helps recover data and systems after failures. Azure Backup and Azure Site Recovery help prevent data loss and restore services.

34. List and describe the advantages and disadvantages of the Azure Shared Responsibility Model.

Answer:

Advantages

1. **Enhanced Security** – Security is improved because both Azure and the customer share the responsibility of protecting cloud resources.
2. **Cost Reduction** – Organizations save money because Azure manages the cloud infrastructure and maintenance.
3. **Business Focus** – Organizations can focus on their core business activities while Azure manages the infrastructure.

Disadvantages

1. Responsibility Confusion – Customers may not clearly understand which security tasks are their responsibility.

2. **Limited Control** – Customers cannot directly control Azure's physical servers and infrastructure.

35. What are the impacts of data loss or unauthorized access in cloud environments?

Answer:

Data loss means data is accidentally deleted, damaged, or destroyed. Unauthorized access means someone accesses data or systems without permission.

Impacts:

- **Financial Loss:** Organizations may lose money due to cyberattacks, data theft, and recovery costs.
- **Legal Issues:** Organizations may face legal problems and fines if they fail to protect customer data.
- **Operational Disruption:** Security problems can stop business activities and important services.
- **Loss of Customer Trust:** Customers may lose trust if their personal or financial information is exposed.

36. Describe Security Posture Assessment and explain its functions in cloud governance.

Answer:

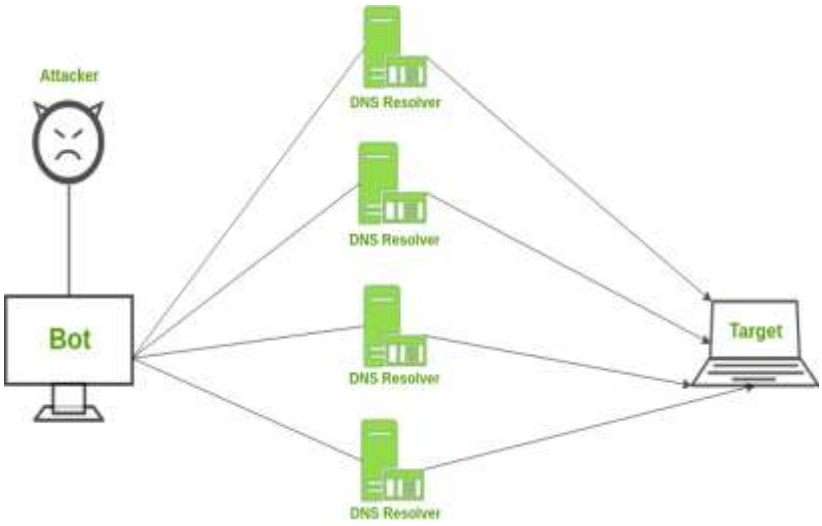
Security Posture Assessment is the process of checking cloud security to find risks, follow security rules, and improve security.

Key Functions

1. **Identify Vulnerabilities:** Find weak passwords, incorrect permissions, outdated software, and unprotected data.
2. **Enforce Security Policies:** Ensure strong passwords, MFA, access control, encryption, and regular software updates.
3. **Strengthen Cloud Governance:** Monitor user activities, identify suspicious actions, and prevent unauthorized access.
4. **Continuous Monitoring:** Regularly check cloud security, identify problems, and suggest ways to improve security.

37. Describe the architecture of a DDoS attack with a suitable example.

Answer:



A Distributed Denial-of-Service (DDoS) attack is a cyberattack in which multiple compromised devices send a large amount of traffic to a target server, website, or network. This overloads the system, making it slow or unavailable to legitimate users.

Example:

A hacker sends a huge number of fake requests from many computers to an online shopping website. This makes the website slow or crash, so real customers cannot access it.

38. Describe the importance of Data Encryption with an example.

Answer:

Data encryption converts readable data into an unreadable format to protect it from unauthorized access. It protects data when stored (data at rest) and when transferred over a network (data in transit).

Encryption is important because even if an attacker gets the data, they cannot read it without the correct decryption key.

Example: A bank stores customer account details in Azure Storage. Encryption protects the information from unauthorized people, even if they gain access to the stored data.

39. Identify the key components of the Azure Security Framework.

Answer:

1. Identity and Access Management (IAM)

IAM controls who can access Azure resources. Azure uses Microsoft Entra ID with MFA, RBAC, SSO, and Conditional Access to provide secure access.

2. Data Encryption

Data Encryption converts readable data into an unreadable format to keep it safe.

3. Monitoring and Threat Detection

Monitoring and Threat Detection continuously checks Azure resources for security threats. It detects suspicious activities and sends alerts to administrators.

4. Network Security and Firewalls

Network Security and Firewalls protect Azure resources from hackers and unwanted network traffic

5. Backup and Disaster Recovery

It helps recover data and systems after failures. Azure Backup and Azure Site Recovery help prevent data loss and restore services.

40. Describe IaaS, PaaS, and SaaS.

Answer:

1. IaaS (Infrastructure as a Service)

In IaaS**,** Azure provides virtual machines, storage, and networking. The customer manages the operating system, applications, and security.

Example: A bank uses an Azure Virtual Machine to run its banking application.

2. PaaS (Platform as a Service)

In PaaS**,** Azure manages the infrastructure, operating system, and platform. The customer develops and manages the application and its data.

Example: College develops an online attendance and examination management system.

3. SaaS (Software as a Service)

In SaaS**,** Azure manages the application, infrastructure, updates, and maintenance. The customer only uses the software and manages user accounts and access.

Example: An organization uses Microsoft 365 for email, Word, Excel, and other office applications.

MODULE 2

IDENTITY AND ACCESS MANAGEMENT IN AZURE

Objective Type Questions(1marks)

1. What is Microsoft Entra ID?

A) Identity management service

B) Storage service

C) Backup service

D) Database service

Answer: A) Identity management service

2. Which process verifies a user's identity?

A) Azure storage

B) Authentication

C) Encryption

D) Monitoring

Answer: B) Authentication

3. Which process decides what a user can access?

A) Authentication

B) Backup

C) Authorization

D) Encryption

Answer: C) Authorization

4. Which Azure feature manages privileged administrator access?

A) Azure Monitor

B) Azure Storage

C) Privileged Identity Management (PIM)

D) Azure Backup

Answer: C) Privileged Identity Management (PIM)

5. What is Single Sign-On (SSO)?

A) Logging in once to access multiple applications

B) Encrypting files

C) Filtering network traffic

D) Monitoring cloud resources

Answer: A) Logging in once to access multiple applications

6. Which of the following manages identities and access?

A) Virtual machines

B) Identity and Access Management (IAM)

C) Storage capacity

D) Network cables

Answer: B) Identity and Access Management (IAM)

7. Which Azure feature assigns permissions through roles?

A) Azure Monitor

B) Azure Backup

C) Azure RBAC

D) Azure DNS

Answer: C) Azure RBAC

8. Which feature requires more than one verification method?

A) SSO

B) RBAC

C) PIM

D) MFA

Answer: D) MFA

9. What is an Azure subscription?

A) A user account

B) A resource and billing boundary

C) A virtual network

D) A storage disk

Answer: B) A resource and billing boundary

10. What is an Azure Resource Group?

A) A user directory

B) A network protocol

C) A container for related resources

D) A backup file

Answer: C) A container for related resources

11. What is the main idea of the Principle of Least Privilege?

A) Give maximum permissions

B) Give only required permissions

C) Give all users admin rights

D) Remove all permissions

Answer: B) Give only required permissions

12. How does authorization relate to access control in Azure?

A) It verifies the user's identity

B) It decides allowed actions

C) It encrypts data

D) It creates backups

Answer: B) It decides allowed actions

13. How does Conditional Access control access to cloud resources?

A) It removes passwords

B) It creates virtual machines

C) It allows or blocks access based on conditions

D) It manages billing

Answer: C) It allows or blocks access based on conditions

14. How does RBAC support the Principle of Least Privilege?

A) By assigning only required role permissions

B) By giving full admin access

C) By removing authentication

D) By sharing passwords

Answer: A) By assigning only required role permissions

15. What is the main purpose of PIM in Azure?

A) Store backup data

B) Provide temporary privileged access

C) Manage network traffic

D) Create subscriptions

Answer: B) Provide temporary privileged access

16. How does authentication support IAM?

A) It verifies the user's identity

B) It assigns storage capacity

C) It encrypts databases

D) It creates networks

Answer: A) It verifies the user's identity

17. How does Conditional Access relate to the Zero Trust model?

A) It trusts all internal requests

B) It disables MFA

C) It verifies access before allowing it

D) It allows every request

Answer: C) It verifies access before allowing it

18. How does Microsoft Entra ID support identity management?

A) It stores VM disks

B) It monitors bandwidth

C) It manages identities and authentication

D) It creates backups

Answer: C) It manages identities and authentication

19. How does a Resource Group help manage Azure resources?

A) It organizes related resources

B) It authenticates users

C) It encrypts passwords

D) It replaces a subscription

Answer: A) It organizes related resources

20. How does an Azure subscription support resource management?

A) It stores only passwords

B) It provides a management and billing boundary

C) It replaces Microsoft Entra ID

D) It works only as a firewall

Answer: B) It provides a management and billing boundary

Short Answer Questions (3marks)

21. Explain in your own words the role of Microsoft Entra ID in Azure.

Answer:

- Microsoft Entra ID is Azure's identity and access management service. It stores information about every user, group, and application in one place.
- It checks who a person is when they try to log in, and this step is called authentication.
- After login, it decides what that person is allowed to do, and this step is called authorization.

22. How does Multi-Factor Authentication (MFA) relate to login security in Microsoft Azure?

Answer:

MFA means a user must prove their identity in more than one way. It does not depend on a password alone. The first way is something the user knows, like a password.

The second way is something the user has, like an OTP sent to their phone.

Even if a password is stolen, the hacker still needs the second proof to log in. This makes the login process much safer in Azure.

23. Illustrate the relationship between Azure Subscription, Resource Group, and Azure Resources with an example.

Answer:

1. Azure Subscription: An Azure Subscription is a container used to manage billing, access, and Azure resources**.**

2. Resource Group: A Resource Group is a container inside a subscription used to organize related resources**.**

3. Azure Resources: Azure Resources are the actual cloud services like Virtual Machines, Storage Accounts and SQL Databases etc**.**

Example:

A company has an Azure Subscription called Company-Azure. Inside it, a Resource Group called HR-System contains resources like Virtual Machine, Database, and Storage Account to run the HR application.

24. How does Identity and Access Management (IAM) help in securing organizational resources?

Answer:

IAM is a system that decides who can access cloud resources and what they are allowed to do. In Azure, IAM is mainly handled by Microsoft Entra ID.

Features

1. Multi-Factor Authentication (MFA)
2. Role-Based Access Control (RBAC)
3. Single Sign-On (SSO)
4. Conditional Access Policies

25. Illustrate the working process of Conditional Access using a simple login scenario.

Answer:

1. **User Sign-in:** The user signs in to an Azure application using their credentials.
2. **Entra ID Evaluation:** Microsoft Entra ID evaluates the sign-in by checking conditions such as the user's identity, device, location, application, etc.
3. **Access Decision:** If the conditions are satisfied, access is granted. If the sign-in is considered risky, Conditional Access requires Multi-Factor Authentication (MFA) or blocks access.
4. **Continuous Monitoring:** After access is granted, Azure continuously monitors the user's session to detect suspicious activity.

26. Illustrate the Just-In-Time (JIT) access process used by PIM in Azure.

Answer:

Just-In-Time (JIT) access in Azure PIM provides temporary administrator access only when needed, instead of permanent access.

Steps:

1. The user requests the privileged role and gives a reason.

2. The user completes MFA and gets approval if required.

3. Azure activates the role for a limited time and removes it automatically after the time ends.

27. Summarize the working process of Role-Based Access Control (RBAC) in Azure.

Answer:

1. Define and Assign Roles: The administrator creates RBAC roles and assigns the appropriate role to the user.

2. Authenticate and Evaluate: The user signs in, and Azure authenticates the user. RBAC checks whether the assigned role has permission to access the requested resource.

3. Grant or Deny Access: If the user has the required permissions, access is granted. Otherwise, access is denied.

28. Illustrate the Privileged Identity Management (PIM) workflow with a simple example.

Answer:

- The user requests the privileged role and gives a reason.
- The user completes MFA and gets approval if required.
- Azure activates the role for a limited time and removes it automatically after the time ends.
- Example: Student requests VM Admin access, completes MFA, gets 1-hour access, uses the VM, and Azure removes access after 1 hour**.**

29. Summarize how Single Sign-On (SSO) improves user productivity.

Answer:

- SSO lets a user log in only once. After that, they can use many apps, like Outlook, Teams, and SharePoint, without logging in again. This saves time for the user.
- It also reduces password fatigue, so users do not reuse weak passwords.
- The security team still keeps full control through Microsoft Entra ID.

30. Illustrate the built-in Azure RBAC roles .

Answer:

1. Reader can only view resources, and they cannot make any changes.
2. Contributor can create, change, and delete resources, but they cannot manage other users' access.
3. Owner can do everything Contributor can do, and they can also manage access for other users, which makes Owner the most powerful role.

Long Answer Questions(5marks)

31. Summarize the two core functions of Identity and Access Management (IAM).

Answer:

1. Authentication

Authentication verifies the identity of a user before granting access. It confirms that the user is who they claim to be by using credentials such as a username and password, and may also require Multi-Factor Authentication (MFA), such as an OTP or fingerprint

2. Authorization

Authorization takes place after successful authentication. It determines what actions the authenticated user is allowed to perform, such as viewing, creating, modifying, or deleting resources. Permissions are assigned using mechanisms such as Role-Based Access Control (RBAC).

32. Explain in your own words Multi-Factor Authentication (MFA) with an example.

Answer:

MFA means a user must prove their identity in more than one way. It does not depend on a password alone. The three authentication factors are:

1. A password or PIN.
2. An OTP sent to a mobile phone or an approval through the Microsoft Authenticator app.
3. A biometric method such as a fingerprint or face scan.

Example:

A bank employee signs in to the Azure portal. First, they enter their username and password. Next, they approve a notification in the Microsoft Authenticator app (or enter an OTP sent to their phone). Only after both steps are completed is access granted.

33. Explain the role of RBAC in securing enterprise resources.

Answer:

1. **Blocks Unauthorized Access:** RBAC prevents users from accessing resources that are not allowed for their role.
2. **Reduces Insider Threats:** RBAC gives users only the required permissions, reducing misuse of access.
3. **Supports Compliance Audits:** RBAC keeps access records that help meet standards like **ISO 27001, GDPR, HIPAA, and PCI DSS.**
4. **Simplifies Administration:** RBAC makes permission management easier by assigning access based on roles instead of individual users.

34. Summarize the Management Group level and Subscription level in Azure RBAC with suitable examples.

Answer:

1. Management Group Level (Highest Level)

This is the broadest scope, controlling all subscriptions under it.

For example, ABC Company's Head Office assigns the Security Administrator role at this level, A Security Admin can manage security for India, USA, and UK subscriptions.

2. Subscription Level

This controls one entire subscription and everything inside it.

For example, the India IT Manager gives Ravi Contributor access at the subscription level**,** Ravi can manage all resources in the India subscription but not USA subscription.

35. Explain in your own words the scope (levels) of Role-Based Access Control (RBAC) in Microsoft Azure.

Answer:

1. Management Group Level (Highest Level)

This is the broadest scope, controlling all subscriptions under it.

For example, ABC Company's Head Office assigns the Security Administrator role at this level, A Security Admin can manage security for India, USA, and UK subscriptions.

2. Subscription Level

This controls one entire subscription and everything inside it.

For example, the India IT Manager gives Ravi Contributor access at the subscription level, Ravi can manage all resources in the India subscription but not USA subscription.

3. Resource Group Level:

This controls one specific resource group. Example: Rithu can manage only Sales-RG resources.

4. Resource Level (Lowest Level):

This controls only one specific resource**.** Example: Kiran can manage only Sales-VM and not other resources.

36. Illustrate HIPAA and PCI DSS compliance standards with suitable examples.

Answer:

HIPAA (Health Insurance Portability and Accountability Act):

HIPAA protects patients' health information. It ensures that only authorized people can access medical records.

Example: A hospital stores patient records in Azure. Before accessing the records, the system verifies that the user is an authorized doctor or nurse. Unauthorized users cannot view the patient's information.

PCI DSS (Payment Card Industry Data Security Standard):

PCI DSS protects credit and debit card information during payment transactions.

Example: An online shopping website accepts card payments. During checkout, the customer's card details are encrypted and protected, ensuring secure payment and preventing unauthorized access.

37. Illustrate Conditional Access and explain how it supports the Zero Trust security model.

Answer:

1. **Never Trust, Always Verify:** Always check users and devices before allowing access.
2. **Verify the User:** Check the user's identity, device, and location before allowing access.
3. **Least Privilege Access:** Give users only the access they need.
4. **Continuous Monitoring and Response:** Monitor sign-ins and block risky access or ask for MFA.
5. **Adaptive Risk-Based Access:** Adjust access based on the risk level.

38. Summarize the advantages and limitations of IAM in Microsoft Azure.

Answer:

Advantages of IAM in Azure

1. **Centralized Control:** IAM allows managing users, roles, and permissions from one place.
2. **Better Security:** IAM provides authentication, authorization, and monitoring to protect resources.
3. **Easy Access:** SSO allows users to access multiple applications with a single login.

Disadvantages of IAM in Azure

1. **Complex Setup:** Setting up IAM requires proper planning and technical knowledge.
2. **Misconfiguration Risk:** Wrong IAM settings can cause security issues or access problems.

39. Illustrate Privileged Identity Management (PIM) in Azure, from Request to Revocation.

Answer:

1. **User Becomes Eligible:** An administrator makes the user eligible for a privileged Azure role instead of assigning it permanently.
2. **User Requests Activation:** The user requests to activate the privileged role and provides a business reason.
3. **Azure Verifies the Request**: Azure checks the request and may require approval and Multi-Factor Authentication (MFA) before granting access.
4. **Azure Activates the Role:** Azure activates the privileged role for a limited time (such as 1 hour or 4 hours).
5. **Azure Logs Activities and Removes Access**: Azure records all user activities for auditing and automatically removes the privileged role when the activation time ends.

40. Summarize the objectives of Conditional Access in Microsoft Azure.

Answer:

1. **Secure Cloud Resources:** Conditional Access protects cloud resources by verifying every sign-in before granting access.
2. **Enforce Zero Trust**: Conditional Access verifies the user's identity, device, location, and risk level before allowing access.
3. **Apply Security Policies:** Conditional Access allows access only to authorized users and secure devices.
4. **Meet Compliance Requirements:** Conditional Access helps organizations follow security rules and standards like GDPR and ISO 27001

MODULE III

NETWORK SECURITY IN AZURE

Objective type questions (1 Marks)

1. What is the main purpose of an Azure VNet?

A) Provide an isolated network

B) Store backups

C) Manage billing

D) Create user accounts

Answer: A) Provide an isolated network

2. What is the main role of Azure Firewall?

A) Store files

B) Centralize network traffic control

C) Manage identities

D) Create databases

Answer: B) Centralize network traffic control

3. What is the main purpose of an NSG?

A) Manage billing

B) Create users

C) Control network traffic with security rules

D) Store backups

Answer: C) Control network traffic with security rules

4. How does an NSG filter network traffic?

A) By user roles

B) By IP addresses, ports, and protocols

C) By storage keys

D) By database tables

Answer: B) By IP addresses, ports, and protocols

5. How does an Azure VNet help secure Azure resources?

A) By grouping resources in a private network

B) By making resources public

C) By disabling security rules

D) By removing subnets

Answer: A) By grouping resources in a private network

6. How does an NSG control inbound and outbound traffic?

A) By using security rules

B) By creating subscriptions

C) By storing passwords

D) By managing backups

Answer: A) By using security rules

7. How does Azure VPN Gateway support secure connectivity?

A) By storing files

B) By creating an encrypted connection

C) By managing user roles

D) By monitoring applications

Answer: B) By creating an encrypted connection

8. How does an NSG use IP addresses, ports, and protocols?

A) To assign user roles

B) To filter network traffic

C) To store keys

D) To manage billing

Answer: B) To filter network traffic

9. What is the main purpose of Azure ExpressRoute?

A) Provide a dedicated private connection

B) Create security rules

C) Manage VMs

D) Store backups

Answer: A) Provide a dedicated private connection

10. How does a subnet relate to a VNet?

A) It replaces the VNet

B) It divides the VNet into smaller networks

C) It manages billing

D) It stores backup files

Answer: B) It divides the VNet into smaller networks

11. How would you use a VNet to secure Azure resources?

A) Expose resources publicly

B) Disable segmentation

C) Isolate resources in a VNet

D) Remove network rules

Answer: C) Isolate resources in a VNet

12. What approach would you use to protect an Azure application from DDoS attacks?

A) Disable monitoring

B) Enable Azure DDoS Protection

C) Remove the VNet

D) Allow all traffic

Answer: B) Enable Azure DDoS Protection

13. How would you use an NSG to protect a VM?

A) Open every port

B) Remove the NSG

C) Allow only authorized traffic

D) Make the VM public

Answer: C) Allow only authorized traffic

14. What would you do to connect an on-premises network securely to Azure?

A) Use an unencrypted link

B) Use Azure DNS

C) Configure Azure VPN Gateway

D) Disable encryption

Answer: C) Configure Azure VPN Gateway

15. What approach would you use to provide private access from a VNet to Azure Storage?

A) Make the storage account publicly accessible

B) Configure a Private Endpoint for the storage account

C) Disable authentication for the storage account

D) Remove the VNet configuration

Answer: B) Configure a Private Endpoint for the storage account

16. What approach would you use for a dedicated private Azure connection?

A) Use Azure ExpressRoute

B) Use Azure Backup

C) Use Azure DNS

D) Use an NSG

Answer: A) Use Azure ExpressRoute

17. How would you secure communication between Azure VMs?

A) Connect them directly to the public Internet

B) Disable network rules

C) Use a VNet with NSG rules

D) Remove the subnet

Answer: C) Use a VNet with NSG rules

18. What would you do to block unauthorized inbound traffic?

A) Allow all traffic

B) Disable the NSG

C) Configure NSG rules

D) Remove authentication

Answer: C) Configure NSG rules

19. How would you use Azure Firewall to secure VM traffic?

A) Allow unrestricted traffic

B) Apply firewall rules to filter traffic

C) Disable the firewall

D) Remove the VNet

Answer: B) Apply firewall rules to filter traffic

20. What approach would you use to improve Azure network security?

A) Use passwords only

B) Disable monitoring

C) Combine VNet, NSG, Firewall, and DDoS Protection

D) Make all resources public

Answer: C) Combine VNet, NSG, Firewall, and DDoS Protection

Short Answer Questions (3 Marks)

21. How would you use a Virtual Network (VNet) to secure Azure resources?

Answer:

1. Create a VNet to provide a private network for Azure resources.
2. Divide the VNet into subnets to separate web servers, application servers, and databases.
3. Apply NSG rules to allow required traffic and block unauthorized traffic.

22. What approach would you use to protect an Azure application from DDoS attacks?

Answer:

Enable Azure DDoS Protection to protect the application from DDoS attacks. It detects when too many requests suddenly reach the application and filters harmful traffic. This helps genuine users access the application without interruption.

23. How would you configure NSG rules to block unauthorized traffic to an Azure virtual machine?

Answer:

1. Create an NSG: Create a Network Security Group in Azure to control traffic to the VM.
2. Configure security rules: Set rules to Allow trusted traffic and Deny unauthorized traffic.
3. Associate the NSG: Attach the NSG to the VM's subnet or network interface to apply the security rules.

24. What approach would you use instead of a VPN Gateway for a dedicated connection to Azure?

Answer:

- **Use Azure ExpressRoute:** Connect the organization's network to Azure using Azure ExpressRoute.
- **Provide a private connection:** It connects to Azure without using the public internet.
- **Ensure reliable performance:** It provides reliable network performance and low latency.

25. How would you secure Azure resources by using a Virtual Network (VNet)?

Answer:

1. Create a VNet to provide a private network for Azure resources.
2. Divide the VNet into subnets to separate web servers, application servers, and databases.
3. Apply NSG rules to allow required traffic and block unauthorized traffic.

26. How would you protect a web server using an NSG?

Answer:

Create an NSG and associate it with the web server's subnet or network interface. Configure the NSG to allow web traffic through ports 80 (HTTP) and 443 (HTTPS). Finally, use security rules to block unauthorized traffic and protect the web server.

27. Write the steps to block unauthorized traffic using NSG rules for an Azure virtual machine.

Answer:

1. Create an NSG: Create a Network Security Group in Azure to control traffic to the VM.
2. Configure security rules: Set rules to Allow trusted traffic and Deny unauthorized traffic.
3. Associate the NSG: Attach the NSG to the VM's subnet or network interface to apply the security rules.

28. How would you apply the three types of firewall rules in Azure Firewall?

Answer:

1. **Network rules:** Filter traffic using IP addresses, ports, and protocols to allow or block connections.
2. **Application rules:** Control web traffic using domain names and URLs to allow trusted websites and block unwanted websites.
3. **NAT rules:** Map private IP addresses to public IP addresses to allow communication with outside networks.

29. How would you solve unauthorized network access using Azure Firewall?

Answer:

Deploy Azure Firewall: Create Azure Firewall in a separate subnet within the VNet.

Configure rules: Allow trusted traffic and block unauthorized traffic.

Monitor traffic: Check network traffic and update firewall rules when needed.

30. How would you use a Private Endpoint to protect Azure Storage?

Answer:

Step 1: Create a Private Endpoint for the Azure Storage account to provide private access.

Step 2: Azure assigns a private IP address to access the storage account through the VNet.

Step 3: Disable public access to protect the storage account from unauthorized access.

Long Answer Questions (5 Marks Each)

31. Explain the concept of an Azure Virtual Network (VNet).

Answer:

An Azure Virtual Network (VNet) is a private network in Azure that allows resources to communicate securely.

1. **Private network:** Resources communicate using private IP addresses.
2. **Isolation:** It keeps resources separate from other networks.
3. **Subnets:** A VNet can be divided into smaller networks called subnets.
4. **Security:** NSG rules control incoming and outgoing network traffic.
5. **Connectivity:** A VNet can connect to an organization's local network using a VPN Gateway.

32. Illustrate the working process of Network Security Groups (NSGs).

Answer:

A Network Security Group (NSG) is a collection of rules that allows or blocks network traffic in Azure.

Working Process of NSG

Step 1: Create an NSG - First, create an NSG in Azure. It acts like a security guard for the network.

Step 2: Create Rules- Set rules to decide which traffic to allow and which traffic to block.

Step 3: Associate the NSG- Attach the NSG to the VM's subnet or network interface so that its rules can control traffic.

Step 4: Check Traffic-When traffic comes in or goes out, Azure checks the NSG rules.

Step 5: Allow or Block- If the rule allows the traffic, it passes through. If the rule denies the traffic, it is blocked.

33. Explain the difference between inbound and outbound network traffic in Azure.

Answer:

Inbound traffic	Outbound traffic
1. It refers to data coming into an azure resource.	1. It refers to data going out of an Azure resource.
2. It controls incoming requests.	2. It controls outgoing requests.
3. Traffic can come from the internet or another network.	3. Traffic can go to the internet or another system.
4. NSG rules decide which incoming traffic is allowed or denied.	4. NSG rules decide which outgoing traffic is allowed or denied.
5. Example: a user accessing a website hosted on an azure server.	5. Example: An Azure virtual machine connecting to an external website.

34. Explain Azure Virtual Network (VNet) with its working process.

Answer:

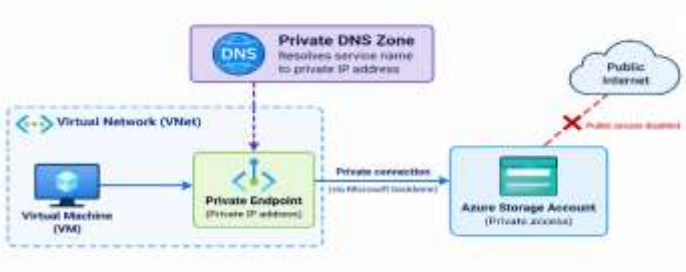
An Azure Virtual Network (VNet) is a private network in Azure that allows resources to communicate securely.

1. **Private network:** Resources communicate using private IP addresses.
2. **Isolation:** It keeps resources separate from other networks.

3. **Subnets:** A VNet can be divided into smaller networks called subnets.
4. **Security:** NSG rules control incoming and outgoing network traffic.
5. **Connectivity:** A VNet can connect to an organization's local network using a VPN Gateway.

35. Explain the architecture of a Private Endpoint in Azure.

Answer:



A Private Endpoint provides secure private access to an Azure service using a private IP address.

1. **Create Private Endpoint:** Connect the VNet to an Azure service.
2. **Private IP:** Azure assigns a private IP address.
3. **DNS Resolution:** DNS maps the service name to the private IP.
4. **Private Communication:** Data travels through Azure's internal network.
5. **Security:** Public access can be disabled to prevent unauthorized access.

Example: A VM securely accesses an Azure Storage account through a Private Endpoint.

36. How does inbound network traffic differ from outbound network traffic in Azure?

Answer:

Inbound traffic	Outbound traffic
1. It refers to data coming into an azure resource.	1. It refers to data going out of an Azure resource.
2. It controls incoming requests.	2. It controls outgoing requests.

3. Traffic can come from the internet or another network.	3. Traffic can go to the internet or another system.
4. NSG rules decide which incoming traffic is allowed or denied.	4. NSG rules decide which outgoing traffic is allowed or denied.
5. Example: a user accessing a website hosted on an azure server.	5. Example: An Azure virtual machine connecting to an external website.

37. Illustrate the working process of Azure VPN Gateway.

Answer:

Azure VPN Gateway provides an encrypted connection between an organization's local network and an Azure VNet through the public internet.

Working process:

1. **Configure the gateway:** Set up a VPN Gateway for the Azure VNet.
2. **Connect the local network:** Configure the VPN device at the organization's location.
3. **Create a secure tunnel:** The gateway establishes an encrypted connection using IPsec.
4. **Transfer data securely:** Data travels through the encrypted tunnel over the internet.
5. **Access Azure resources:** The local network can communicate securely with resources in the Azure VNet.

38. Summarize the Azure networking services used to secure cloud resources.

Answer:

Azure provides several networking services to protect cloud resources.

1. **Virtual Network (VNet)**: Provides a private network for Azure resources.
2. **Network Security Group (NSG):** Filters inbound and outbound traffic using security rules.
3. **Azure Firewall:** Centrally monitors and filters network traffic.
4. **Private Endpoint and VPN Gateway:** Provide private access to Azure services and encrypted connections between local networks and Azure.
5. **DDoS Protection and ExpressRoute:** DDoS Protection helps defend against traffic-flooding attacks, while ExpressRoute provides a dedicated private connection to Azure. These services work together to improve network security.

39. Explain Azure ExpressRoute with suitable examples.

Answer:

Azure ExpressRoute provides a private connection between an organization's local network and Azure without using the public internet.

1. **Private Connection:** Connects the company network to Azure through a private circuit.
2. **Network Isolation:** Data travels through a private route.
3. **Reliable Performance:** Provides stable speed and low delay.
4. **Redundancy:** Provides backup network paths if one path fails.
5. **Cloud Access:** Connects local systems to Azure services.

Example: A hospital uses ExpressRoute to connect its local data centre to Azure securely and reliably.

40. Explain Azure Firewall and its working process in your own words.

Answer:

Azure Firewall is a managed cloud security service that monitors and filters network traffic. It helps protect Azure resources from unauthorized access.

Working Process:

1. **Deploy Firewall:** Create Azure Firewall in a dedicated subnet.
2. **Route Traffic:** Direct network traffic through the firewall.
3. **Check Traffic:** Check traffic using security rules.
4. **Allow or Block:** Allow safe traffic and block unwanted traffic.
5. **Monitor Threats:** Detect and block known malicious sources.

MODULE IV

DATA AND PLATFORM PROTECTION ENCRYPTION

Objective Type Questions (1 Marks)

1. How would you protect data stored in Azure Blob Storage?

A) Disable encryption

B) Enable Azure Storage Service Encryption

C) Make the container public

D) Store data as plain text

Answer: B) Enable Azure Storage Service Encryption

2. How would you protect data moving between a browser and Azure?

A) Use data compression

B) Use HTTP only

C) Use HTTPS/TLS

D) Delete the data

Answer: C) Use HTTPS/TLS

3. How would you securely manage keys, secrets, and certificates in Azure?

A) Use Azure Key Vault

B) Use Azure DNS

C) Use Azure Monitor

D) Use Azure Firewall

Answer: A) Use Azure Key Vault

4. What approach would you use to protect data in Azure SQL Database?

A) Use Azure DNS

B) Use TDE

C) Use a Load Balancer

D) Use an NSG

Answer: B) Use TDE

5. What approach would you use to protect Azure VM disks?

A) Disable encryption

B) Make the disks public

C) Use Azure Disk Encryption

D) Remove disk security

Answer: C) Use Azure Disk Encryption

6. How would you securely manage cryptographic keys in Azure?

A) Email the keys to users

B) Store the keys in a text file

C) Store and manage the keys using Azure Key Vault

D) Hardcode the keys in the application

Answer: C) Store and manage the keys using Azure Key Vault

7. How would you identify vulnerabilities and misconfigurations in Azure workloads?

A) Use Azure DNS

B) Use Microsoft Defender for Cloud

C) Use a Load Balancer

D) Disable alerts

Answer: B) Use Microsoft Defender for Cloud

8. How would you keep passwords and API keys out of application code?

A) Hardcode them

B) Save them in a public file

C) Store them in Azure Key Vault

D) Print them in logs

Answer: C) Store them in Azure Key Vault

9. How would you protect data while it is stored in Azure?

A) Use encryption at rest

B) Use compression only

C) Disable encryption

D) Store it as plain text

Answer: A) Use encryption at rest

10. How would you protect data while it is being transferred?

A) Use plain HTTP

B) Disable TLS

C) Use encryption in transit

D) Use compression only

Answer: C) Use encryption in transit

11. What would you do to protect sensitive data in Azure Storage?

A) Make the container public

B) Enable encryption at rest

C) Remove encryption

D) Disable access control

Answer: B) Enable encryption at rest

12. How would you securely manage application secrets?

A) Store them in source code

B) Email them

C) Store them in Azure Key Vault

D) Put them in a public repository

Answer: C) Store them in Azure Key Vault

13. How would you secure data sent to an Azure web application?

A) Use HTTPS/TLS

B) Use HTTP only

C) Use Disk Encryption

D) Use compression only

Answer: A) Use HTTPS/TLS

14. What approach would you use to protect sensitive VM disks?

A) Leave them unencrypted

B) Apply Azure Disk Encryption

C) Make them public

D) Disable disk security

Answer: B) Apply Azure Disk Encryption

15. What approach would you apply to securely manage passwords and application secrets in Azure?

A) Store them in a configuration file

B) Print them in application logs

C) Store and manage them using Azure Key Vault

D) Share them through email

Answer: C) Store and manage them using Azure Key Vault

16. How would you protect sensitive organizational data at rest?

A) Store it as plain text

B) Disable encryption

C) Use encryption at rest

D) Make it public

Answer: C) Use encryption at rest

17. What would you do to protect information transferred between Azure services?

A) Use unencrypted FTP

B) Disable TLS

C) Use encryption in transit

D) Remove security controls

Answer: C) Use encryption in transit

18. What approach would you use to securely manage certificates for an Azure application?

A) Store them in a shared drive

B) Embed them in source code

C) Use Azure Key Vault

D) Leave them unmanaged

Answer: C) Use Azure Key Vault

19. What approach would you use to prevent API keys from being exposed?

A) Hardcode the keys

B) Share them by email

C) Save them in a plain-text file

D) Store them in Azure Key Vault

Answer: D) Store them in Azure Key Vault

20. What approach would you use to identify security weaknesses in Azure workloads?

A) Disable security alerts

B) Use manual review only

C) Use Azure DNS

D) Use Microsoft Defender for Cloud

Answer: D) Use Microsoft Defender for Cloud

Short Answer Questions (3 Marks)

21. How would you implement encryption at rest for sensitive organizational data stored in Microsoft Azure?

Answer:

Enable encryption at rest to protect sensitive data stored in Azure. Use Azure Storage Service Encryption for storage data, Azure Disk Encryption for virtual machine disks, or TDE for SQL databases. This helps protect stored data from unauthorized access.

22. Demonstrate how encryption in transit protects data when a user accesses an Azure-based application through a web browser.

Answer:

- Use HTTPS/TLS to secure the connection between the browser and the Azure application.
- It encrypts data while it travels over the internet.
- It protects data from attackers and prevents them from reading it.

23. How would you use Azure Key Vault to securely manage cryptographic keys, secrets, and certificates for an Azure application?

Answer:

Store cryptographic keys, passwords, API keys, and certificates securely in Azure Key Vault instead of storing them in application code. Use Role-Based Access Control (RBAC) to allow

only authorized users and applications to access them. Rotate keys regularly to maintain security and reduce the risk of unauthorized access.

24. What approach would you use to protect Azure SQL Database data using Transparent Data Encryption (TDE)?

Answer:

Enable Transparent Data Encryption (TDE) for Azure SQL Database to protect stored data. TDE automatically encrypts database files and transaction log files. This helps protect sensitive data from unauthorized access if the database files are accessed directly.

25. How would you apply Azure Disk Encryption to protect operating system and data disks of an Azure virtual machine?

Answer:

- Enable Azure Disk Encryption for the virtual machine's operating system and data disks.
- Manage the encryption keys securely to prevent unauthorized access.
- This helps protect stored data from unauthorized access.

26. How would you secure application passwords, API keys, and connection strings in an Azure-based application using Azure Key Vault?

Answer:

- Store passwords, API keys, and connection strings in Azure Key Vault.
- Allow the application to access them securely using a managed identity.
- Get the secrets whenever needed instead of saving them in the application code.

27. Demonstrate how Microsoft Defender for Cloud can be used to identify vulnerabilities and security misconfigurations in Azure workloads.

Answer:

Microsoft Defender for Cloud monitors Azure resources and identifies security problems and incorrect security settings. It provides recommendations to fix these problems. This helps improve the security of Azure resources.

28. What approach would you use to protect sensitive organizational data stored in Azure using secure storage practices?

Answer:

Enable encryption to protect sensitive data stored in Azure. Use RBAC and ACLs to give access only to authorized users. Use Private Endpoints to connect to Azure Storage through a private network and protect the data from unauthorized access.

29. How would you configure an Azure-based application to protect data both at rest and in transit?

Answer:

Enable encryption at rest to protect data stored in Azure databases, disks, and storage. Use HTTPS/TLS to secure data while it is transferred between the user and the Azure application. Use Azure Key Vault to securely manage encryption keys and application secrets.

30. How would you use cryptographic key management to protect encrypted data in an Azure-based application?

Answer:

Store and manage encryption keys securely in Azure Key Vault instead of storing them in application code. Use Role-Based Access Control (RBAC) to allow only authorized users and applications to access them. Rotate keys regularly to maintain security and reduce the risk of unauthorized access.

Long Answer Questions (5 Marks)

31. How would you design a data protection strategy for a banking application in Azure using encryption, Key Vault, and secure storage?

Answer:

1. **Encryption:** Use encryption at rest to protect customer records and transaction data stored in Azure.
2. **Encryption in Transit:** Use HTTPS/TLS to secure data transferred between customers and the banking application.
3. **Azure Key Vault:** Store encryption keys, passwords, and certificates securely.
4. **Access Control:** Apply RBAC to restrict access to authorized users and use Private Endpoints for secure network connectivity.

5. **Security Monitoring:** Use Microsoft Defender for Cloud to identify security problems and incorrect settings.

32. How would you implement end-to-end data protection for an Azure-based application using encryption at rest and encryption in transit?

Answer:

Encryption at rest:

- Encrypt data stored in Azure databases, virtual machine disks, and storage accounts.
- Use Azure Storage Service Encryption, Disk Encryption, or TDE to protect stored data.

Encryption in transit:

- Use HTTPS/TLS to encrypt data transferred between users and the Azure application.
- Secure communication between Azure services to protect data during transmission.
- Use Azure Key Vault to manage encryption keys securely.

33. What approach would you use to secure sensitive VM disks in Azure using Azure Disk Encryption?

Answer:

Azure Disk Encryption is used to protect the data stored on a virtual machine's operating system and data disks.

- First, enable Azure Disk Encryption on the required disks. It encrypts the stored data and helps prevent unauthorized access.
- Encryption keys must be managed securely, and only authorised users should be allowed to access the virtual machine.
- Microsoft Defender for Cloud can also be used to identify security problems and improve protection.

34. How would you design a secure cryptographic key, secret, and certificate management system using Azure Key Vault for an enterprise application?

Answer:

Azure Key Vault is used to store and manage cryptographic keys, passwords, API keys, tokens, and certificates securely.

- First, store these sensitive items in Key Vault instead of directly in the application code.

- Then, use Role-Based Access Control (RBAC) to allow only authorized users and applications to access them.
- The application can use a managed identity to access Key Vault securely. Regularly update keys and secrets, and monitor access to identify unusual activity.

35. How would you use Microsoft Defender for Cloud to monitor Azure resources for vulnerabilities and misconfigurations?

Answer:

Microsoft Defender for Cloud helps monitor Azure resources and identify security weaknesses.

1. Enable monitoring: Use Microsoft Defender for Cloud to monitor Azure resources.
2. Identify vulnerabilities: Detect weaknesses in applications and workloads.
3. Check configurations: Identify misconfigurations, such as publicly exposed storage.
4. Generate alerts: Notify the security team about security problems.
5. Fix problems: Follow security recommendations and apply the required corrections.

36. How would you design a secure Azure data protection architecture for a university application that stores student records, examination information, and personal details?

Answer:

Use encryption at rest to protect student records, examination marks, and personal details stored in Azure.

Use HTTPS/TLS to protect data during transmission.

Store encryption keys, passwords, and application secrets securely in Azure Key Vault.

Apply RBAC so students can access their own information and authorized staff can access required records.

Use Microsoft Defender for Cloud to identify security vulnerabilities and incorrect settings.

37. How would you implement a secure Azure data protection strategy using encryption, key management, secure storage, and Microsoft Defender for Cloud?

Answer:

A secure data protection strategy uses different security methods to protect information in Azure.

1. **Encryption:** Use encryption at rest and encryption in transit to protect data.
2. **Key management:** Use Azure Key Vault to manage keys, passwords, and certificates.
3. **Secure storage:** Use RBAC, ACLs, and Private Endpoints to control storage access.

4. **Monitoring:** Use Microsoft Defender for Cloud to identify vulnerabilities and misconfigurations.
5. **Compliance:** Use Azure Policy to enforce security rules and Secure Score to track security improvements.

38. What approach would you use to protect sensitive Azure data using encryption, Key Vault, and secure storage?

Answer:

- Use encryption at rest to protect sensitive data stored in Azure and HTTPS/TLS to secure data during transmission.
- Store encryption keys, passwords, and certificates securely in Azure Key Vault.
- Apply Role-Based Access Control (RBAC) to allow only authorized users and applications to access the required resources.
- Use Private Endpoints to provide private network connectivity to Azure Storage.
- These measures help protect sensitive data from unauthorized access.

39. How would you protect an Azure SQL Database using Transparent Data Encryption and secure key management?

Answer:

TDE protects an Azure SQL Database by automatically encrypting its database files and transaction log files.

- First, enable TDE for the database to protect stored data.
- Manage the encryption keys securely to help prevent unauthorized access to the encrypted information.
- Use RBAC to control access to database resources and ensure that only authorized users have the required permissions.
- Microsoft Defender for Cloud can also help identify database security problems and provide recommendations to improve protection.

40. How would you implement secure storage practices for sensitive organisational data in Azure?

Answer:

- Use encryption at rest to protect sensitive data stored in Azure and HTTPS/TLS to secure data during transmission.
- Store encryption keys, passwords, and certificates securely in Azure Key Vault.
- Apply Role-Based Access Control (RBAC) to allow only authorized users and applications to access the required resources.
- Use Private Endpoints to provide private network connectivity to Azure Storage.
- These measures help protect sensitive data from unauthorized access.

MODULE 5

SECURITY MONITORING AND COMPLIANCE

Objective Type Questions (1 Marks)

1. What approach would you use to detect a public Azure Storage misconfiguration?

A) Use Azure DNS

B) Use Microsoft Defender for Cloud

C) Use Azure Load Balancer

D) Use Azure VNet

Answer: B) Use Microsoft Defender for Cloud

2. What approach would you use to analyze collected Azure security logs?

A) Use Azure Storage

B) Use Azure DNS

C) Use Log Analytics

D) Use Azure Load Balancer

Answer: C) Use Log Analytics

3. What would you do if an Azure VM has known security vulnerabilities?

A) Ignore the alert

B) Disable monitoring

C) Apply security patches

D) Delete the VM

Answer: C) Apply security patches

4. How would you respond to repeated failed admin logins from one IP?

A) Ignore the attempts

B) Investigate for a brute-force attack

C) Give more access

D) Disable logging

Answer: B) Investigate for a brute-force attack

5. What approach would you use if a user suddenly accesses data from another country?

A) Treat it as anomalous activity

B) Grant more access

C) Ignore the activity

D) Disable monitoring

Answer: A) Treat it as anomalous activity

6. What approach would you use to detect a malicious file on an Azure VM?

A) Use Azure DNS

B) Use a Load Balancer

C) Use Microsoft Defender for Cloud

D) Use Resource Manager

Answer: C) Use Microsoft Defender for Cloud

7. What approach would you use to detect an unusual increase in login attempts?

A) Delete the logs

B) Use anomaly detection

C) Disable monitoring

D) Compress the files

Answer: B) Use anomaly detection

8. How would you continuously monitor Azure logs and metrics?

A) Use Azure Monitor

B) Use Azure DNS

C) Use Storage Explorer

D) Use a Load Balancer

Answer: A) Use Azure Monitor

9. What approach would you use when patient data is unencrypted and users have excessive permissions?

A) Increase storage capacity

B) Ignore the issue

C) Encrypt the data and reduce excessive permissions

D) Disable auditing

Answer: C) Encrypt the data and reduce excessive permissions

10. What approach would you use to check Azure resources against ISO 27001 and GDPR requirements?

A) Use Azure DNS

B) Use Defender for Cloud compliance tools

C) Use a Load Balancer

D) Use Azure VNet

Answer: B) Use Defender for Cloud compliance tools

11. What approach would you use to secure publicly exposed Azure Storage?

A) Assume it is safe

B) Wait for a complaint

C) Detect and correct the misconfiguration

D) Disable monitoring

Answer: C) Detect and correct the misconfiguration

12. What approach would you use for a VM with missing security patches?

A) Leave it outdated

B) Disable monitoring

C) Apply the required patches

D) Delete security logs

Answer: C) Apply the required patches

13. How would you respond to repeated failed admin logins?

A) Allow more attempts

B) Ignore the attempts

C) Identify brute-force activity and apply controls

D) Remove MFA

Answer: C) Identify brute-force activity and apply controls

14. What would you do if a user accesses sensitive resources from an unusual location?

A) Grant automatic access

B) Require additional verification

C) Ignore the activity

D) Disable all alerts

Answer: B) Require additional verification

15. What approach would you use during an Azure security investigation with Azure Monitor?

A) Disable telemetry

B) Delete logs

C) Review logs and metrics

D) Use screenshots only

Answer: C) Review logs and metrics

16. What approach would you use to investigate suspicious security logs?

A) Delete the logs

B) Query them with Log Analytics

C) Send them to an unmonitored server

D) Ignore them

Answer: B) Query them with Log Analytics

17. What would you do if malware starts running on an Azure VM?

A) Leave the VM running

B) Disable monitoring

C) Use Defender for Cloud and isolate the VM

D) Ignore the malware

Answer: C) Use Defender for Cloud and isolate the VM

18. How would you detect unusual login activity?

A) Ignore the pattern

B) Review it once a year

C) Delete authentication logs

D) Use statistical anomaly detection

Answer: D) Use statistical anomaly detection

19. What approach would you use to check ISO 27001 and GDPR compliance?

A) Disable audit logs

B) Assume compliance

C) Use compliance monitoring and reports

D) Check only after an incident

Answer: C) Use compliance monitoring and reports

20. What approach would you use to improve Azure security using Secure Score?

A) Use Secure Score for billing

B) Ignore security recommendations

C) Disable security settings

D) Follow Secure Score recommendations

Answer: D) Follow Secure Score recommendations

Short Answer Questions (3 Marks)

21. How would you use Microsoft Defender for Cloud, Azure Monitor, and Log Analytics to monitor Azure for security threats?

Answer:

1. **Microsoft Defender for Cloud** checks Azure resources for vulnerabilities and security problems and provides a **Secure Score**.
2. **Azure Monitor** monitors the performance and activities of Azure resources.
3. **Log Analytics** collects and analyzes logs to find unusual login and activity information.
   Example: A college student portal running on an Azure VM can be continuously monitored using these tools.

22. How would you generate a compliance report for an organization that must follow ISO 27001 and GDPR?

Answer:

1. The organization selects the required security standards, ISO 27001 and GDPR.
2. Azure automatically checks the security settings. If it finds any problems, it reports them to the organization.
3. The organization fixes the problems and downloads the compliance report from the Azure dashboard.

23. What approach would you use when an Azure virtual machine is found to have outdated software and missing security patches?

Answer:

First, identify the outdated software and missing security patches on the Azure VM. Use Microsoft Defender for Cloud to check for vulnerabilities and other security issues. Update the outdated software and apply the required security patches. Finally, scan the VM again to check for any remaining security issues.

24. How would you solve the problem of repeated failed login attempts against an administrator account using what you learned about threat protection?

Answer:

The Repeated failed login attempts may indicate a Brute-Force Attack, where an attacker tries many passwords to access an account. Security tools detect suspicious activity and alert the security team. The team investigates the alert and blocks the threat to protect the administrator account.

25. What would you do if a user's Azure account suddenly logs in from an unusual country and accesses sensitive resources?

Answer:

Check the sign-in logs to find out why the user logged in from an unusual country. Use Azure Monitor and Log Analytics to investigate the activity. If the login is unauthorized, block the sign-in, reset the password, and inform the security team.

26. Demonstrate how you would respond when malware is detected on an Azure virtual machine.

Answer:

If malware is detected on an Azure virtual machine, identify the affected VM from the security alert. Isolate it from the network to stop the malware from spreading. Remove the malware using security tools, update the system, and scan it again.

27. How would you use anomaly detection techniques to identify unusual activities in an Azure cloud environment?

Answer:

Azure Monitor collects login and activity information from Azure resources. Log Analytics analyzes the logs to identify unusual activities such as unusual logins or large data downloads. Microsoft Defender for Cloud detects security threats and generates alerts for the security team.

28. What approach would you use to investigate a suspicious security incident using Azure Monitor and Log Analytics?

Answer:

1. **Azure Monitor** collects the user's login and activity information.
2. **Log Analytics** analyzes the logs to find unusual login or activity information.
3. If suspicious activity is found, the security team investigates it and takes the necessary action.

Example: If a user logs in from an unusual location, the logs can be checked to determine whether the activity is suspicious.

29. What would you do if a healthcare organization discovers that patient data is not encrypted and a user has excessive permissions?

Enable suitable encryption to protect stored patient data. Review the user's permissions and remove unnecessary access using role-based access control (RBAC). Check the activity logs for possible misuse and regularly review security settings.

30. How would you identify abnormal user activities in an Azure environment using anomaly detection techniques?

Answer: Azure Monitor collects login and activity information from Azure resources. Log Analytics analyzes the logs to identify unusual activities such as unusual logins or large data downloads. Microsoft Defender for Cloud detects security threats and generates alerts for the security team.

Long Answer Questions (5 Marks)

31. How would you use Microsoft Defender for Cloud, Azure Monitor, and Log Analytics for continuous security monitoring?

Answer:

- Microsoft Defender for Cloud checks for vulnerabilities and security problems.
- It gives a Secure Score for security.
- Azure Monitor checks performance and resource activity.
- Log Analytics stores logs and checks login and activity information.
- Together, they monitor Azure and find security problems.

Scenario:

A college student portal runs on an Azure VM. These tools monitor the VM, find security problems, and check login and activity information.

32. What approach would you apply to protect an Azure VM when malware is detected?

1. Check the security alert to identify the affected virtual machine and the detected malware.
2. Isolate the VM from the network to reduce the risk of the malware spreading.
3. Investigate the logs to understand the possible impact of the malware.
4. Remove the malware using trusted security tools or restore the VM from a clean backup if necessary.
5. Install security updates, scan the VM again, and monitor it for further suspicious activities.

33. What approach would you use to prevent a brute-force attack on an administrator account in Azure?

Answer:

Enable MFA to provide extra security, so a password alone is not enough to access the account. Use a strong and unique password for the administrator account. Apply Conditional Access policies to restrict risky sign-ins. Give administrator permissions only to users who need them for their work. Monitor failed login attempts and check security alerts regularly to identify suspicious activities.

Example: If someone tries to access the administrator account using an incorrect password, MFA provides an extra layer of security.

34. What would you do if a valid Azure user account suddenly accesses sensitive resources from an unusual location and downloads a large amount of data?

Answer:

First, Azure Monitor collects the user's login and activity information.

Then, Log Analytics checks the logs to find unusual login activity**.**

If it finds a login from an unusual location and a large amount of data being downloaded**,** it is treated as suspicious activity.

Microsoft Defender for Cloud detects the threat and sends an alert to the administrator.

The administrator can then investigate and take necessary action**.**

35. Demonstrate how you would apply different anomaly detection techniques to identify suspicious activities in an Azure cloud environment.

Answer:

Azure Monitor collects activity and login information from Azure resources. Then, Log Analytics checks the logs to find unusual activities.

Different anomalies can be identified, such as:

- Unusual location: Login from an unknown location.
- Unusual time: Login at an unusual time.
- Unusual activity: A user performs activities different from normal.
- Large data download: A user suddenly downloads a large amount of data.

When suspicious activity is found, Microsoft Defender for Cloud can generate an alert. The security team can investigate the activity and take suitable action.

36. Design a continuous compliance monitoring system for an organization that needs to follow ISO 27001 and GDPR requirements.

Answer:

- Select the required security standards, such as ISO 27001 and GDPR.
- Use Microsoft Defender for Cloud to check security recommendations and compliance status.
- Use Azure Policy to check whether Azure resources follow security rules.
- Use Azure Monitor and Log Analytics to collect and check logs for security problems.
- Prepare compliance reports regularly and check whether identified problems are fixed.

37. How would you use Microsoft Defender for Cloud to improve the security and compliance posture of an Azure organization? Explain.

Answer:

- Enable Microsoft Defender for Cloud to find security problems and vulnerabilities in Azure resources.
- Check the Secure Score to understand the organization's security level.
- Follow security recommendations to fix outdated software and unsafe settings.
- Use the compliance dashboard to check whether the organization follows standards like ISO 27001 and GDPR.
- Review security alerts regularly and take action to protect Azure resources.

38. How would you solve a situation where security teams receive a large number of alerts from different Azure workloads and important threats may be missed?

Answer:

1. Azure Monitor collects activity and event information from Azure resources.
2. Log Analytics checks the logs to find important or unusual activities.
3. Repeated or unnecessary alerts are identified to reduce confusion.
4. Important alerts are given more attention, especially alerts related to sensitive resources.
5. Microsoft Defender for Cloud identifies security risks and provides security alerts.
6. The security team checks the important alerts first and takes the necessary action.

39. What approach would you use when a healthcare organization discovers that patient information is stored without encryption and users have excessive permissions?

Answer:

- Encrypt patient data stored in Azure to prevent unauthorized people from accessing it.
- Review user permissions and remove access that is not required for their work.
- Apply Role-Based Access Control (RBAC) to ensure that users can access only the data and resources needed for their jobs.
- Use Azure Key Vault to securely store and manage encryption keys.
- Monitor activity logs to identify unauthorized access and regularly review security settings to protect patient information.

40. What would happen if an organization did not continuously monitor its Azure workloads for vulnerabilities, misconfigurations, and suspicious activities? Suggest suitable security measures.

Answer:

The lack of continuous monitoring can cause security problems to go unnoticed. Old software may have security problems, unsafe settings may expose resources, and unusual activities may not be detected.

Security measures:

1. Use Microsoft Defender for Cloud to find security risks.
2. Use Azure Monitor to collect activity information.
3. Use Log Analytics to check logs for unusual activities.
4. Use Azure Policy to check whether settings follow security rules.
5. Check Secure Score regularly to know the security level.